



1.
(a).

$$\begin{aligned}
 P(\bar{A} \cap \bar{B}) &= P(\bar{B}) - P(A \cap \bar{B}) \quad \text{total probability} \\
 &= P(\bar{B}) - (P(A) - P(A \cap B)) \\
 &= P(\bar{B}) - (P(A) - P(A)P(B)) \quad \text{independent.} \\
 &= P(\bar{B}) - P(A)(1 - P(B)) \\
 &= P(\bar{B}) \cdot P(\bar{A}). \quad \text{proved.}
 \end{aligned}$$

$$\begin{aligned}
 P(\bar{A} \cap \bar{B}) &= P(\overline{A \cup B}) \quad \text{De Morgan's Law.} \\
 &= 1 - P(A \cup B) \quad \text{容斥} \\
 &= 1 - P(A) - P(B) + P(A)P(B) \\
 &= P(\bar{A})P(\bar{B}).
 \end{aligned}$$

(b). According to (a):

$$\begin{aligned}
 P(A \cap \bar{B}) &= P(\bar{B}) - P(\bar{A} \cap \bar{B}) \\
 &= P(\bar{B}) - P(\bar{A}) \cdot P(\bar{B}) \\
 &= P(\bar{B}) \cdot P(A). \quad \text{proved.}
 \end{aligned}$$

(c). we assume roll a dice which has 6 numbers from 1 to 6, that produces a uniform probability space. A is the event that we roll 1.

$$P(A) = \frac{1}{6} \quad P(\bar{A}) = \frac{5}{6}$$

$$P(A \cap A) = 0 \neq P(A) \cdot P(A) = \frac{1}{36} \quad \text{disproved.}$$





(d).

$A=B$, then.

We assume $P(A) = P(B)$, $P(A \cap B) = P(A) = P(B) = P(A) \cdot P(B) = P(A)^2$ (independence).

$$\therefore P(A) \cdot (P(A) - 1) = 0.$$

$$\therefore P(A) = 1 \text{ or } P(A) = 0.$$

When $P(A) = P(B) = 1$ or $P(A) = P(B) = 0$, it's possible that $A=B$ proved.

2.

A is the event a lie detector is reliable.

B is the event the ^{detected} person is guilty.

C is the event the guilty person is chosen.

$$P(A|B) = \frac{4}{5} \quad P(A|\bar{B}) = \frac{9}{10}$$

$$P(C) = \frac{1}{100} = P(B).$$

$$P(A) = \frac{P(A \cap B)}{P(B)} + \frac{P(A \cap \bar{B})}{P(\bar{B})} \rightarrow \text{total probability}$$

$$P(A) = P(A|B) \cdot P(B) + P(A|\bar{B}) \cdot P(\bar{B})$$

$$= \frac{4}{5} \times \frac{1}{100} + \frac{9}{10} \times \frac{99}{100}$$

$$= \frac{8}{1000} + \frac{891}{1000}$$

$$= \frac{899}{1000}$$





3.

(a). A_i is the event the i -th is head.

The flipping coins is mutually independent. And we assume it's uniform.

$$P(\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3 \cap A_4 \cap A_5) + P(\bar{A}_1 \cap A_2 \cap \bar{A}_3 \cap A_4 \cap A_5)$$

$$+ P(A_1 \cap \bar{A}_2 \cap \bar{A}_3 \cap A_4 \cap A_5) = \left(\frac{1}{2}\right)^5 \times 3 = \frac{3}{32}$$

(b).

B is the event that the game ends at fifth dice.

C is the event that there're three heads.

$$P(C) = \frac{3}{32}$$

$$P(C|B) = \frac{P(C \cap B)}{P(B)} = \frac{\frac{3}{32}}{\frac{3}{32}} = \frac{3}{32}$$

(c).

$$3: A_1 \cap A_2 \cap A_3 \quad \bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3 \quad \frac{1}{2^3} \times 2$$

$$4: \bar{A}_1 \cap A_2 \cap A_3 \cap A_4 \quad A_1 \cap \bar{A}_2 \cap \bar{A}_3 \cap \bar{A}_4 \quad \frac{1}{2^4} \times 2$$

$$5: A_1 \cap \bar{A}_2 \cap A_3 \cap A_4 \cap A_5 \quad \bar{A}_1 \cap \bar{A}_2 \cap A_3 \cap A_4 \cap A_5$$

$$6: \left. \begin{array}{l} \text{three heads.} \\ A_1 \cap A_2 \cap \bar{A}_3 \cap A_4 \cap A_5 \cap A_6 \\ A_1 \cap \bar{A}_2 \cap \bar{A}_3 \cap A_4 \cap A_5 \cap A_6 \\ \bar{A}_1 \cap A_2 \cap \bar{A}_3 \cap A_4 \cap A_5 \cap A_6 \end{array} \right\} \begin{array}{l} \text{three tail} \\ A_1 \cap A_2 \cap \bar{A}_3 \cap \bar{A}_4 \cap \bar{A}_5 \\ \bar{A}_1 \cap \bar{A}_2 \cap A_3 \cap \bar{A}_4 \cap \bar{A}_5 \\ \bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3 \cap \bar{A}_4 \cap \bar{A}_5 \end{array}$$

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$$\bar{A}_1 \cap A_2 \cap A_3 \cap \bar{A}_4 \cap \bar{A}_5 \cap \bar{A}_6$$





④

A win when the first flipping is head and the second flipping is tail.

B win when the first flipping is tail and the second flipping is head.

$$P(A) = P(B) = p(1-p).$$

X

⑤ ? ? ?

(a) X

A_i : there're exactly i persons getting right card.

B_i : there're exactly i persons getting wrong card.

$$S = \{(a_1, \dots, a_n), i, j \mid a_i \in \{1, \dots, n-1\}\}.$$

$$P(A_1) = P(B_{n-1})$$

$$P(A_2) = P(B_{n-2})$$

$$P(A_{n-1}) = P(B_1) = 0$$

$$P(A_n) = P(B_0) = \frac{1}{n!}$$

$$A_i \cap A_j = \emptyset$$

$$P(B_2) = \frac{\frac{C_n^2}{n!}}{(n-2)!} = \frac{\frac{n(n-1)}{2}}{n!}$$

$$P(B_n) = 1 - P(A_1 \cup A_2 \cup A_3 \dots \cup A_n)$$

$$\therefore S_n = 1 - \frac{1}{n!}$$

$$= 1 - \sum_{i=1}^n P(A_i)$$

$$S_{n+1} = 1 - \frac{1}{(n+1)!}$$

$$= 1 - \left(\sum_{i=1}^{n-1} P(B_i) + \frac{1}{n!} \right).$$

$$P(B_n) = a_n = \frac{1}{(n-1)!} - \frac{1}{n!}$$

$$a_n = 1 - S_{n-1} - \frac{1}{n!}$$





不是所有无向可重序列都
样本点不可能 是可行的

~~(a).~~

$$\frac{1}{n} \cdot \frac{C_{n-k+n-2}^{n-k}}{C_{n+n-1}^n} \times$$

$$= \frac{C_{2n-k-2}^{n-k}}{C_{2n-1}^n}$$

无向可重序列.
n个类别中可重选 $\frac{n}{k}$ 箱 球. $\rightarrow \binom{n+k-1}{k} = |S|$ 其中有 k 个在 first bin

的样本点: $\binom{n}{k} \binom{2n-k-2}{n-k}$

$$\frac{(2n-k-2)!}{(n-k)! \cdot (n-2)!} \cdot \frac{n!}{(2n-1)!} = \frac{n!}{(n-k)!} \cdot \frac{(n-1)! (2n-k-2)!}{(2n-1)! (n-2)!}$$

$$\frac{\binom{n}{k} \binom{2n-k-1}{n-k}}{\binom{n+k-1}{k}}$$

(b).

$$\sum_{k=\frac{n}{2}}^n \frac{C_{2n-k-2}^{n-k}}{C_{2n-1}^n} \times$$

(c). A_i : i-th bin has at least half of the balls.

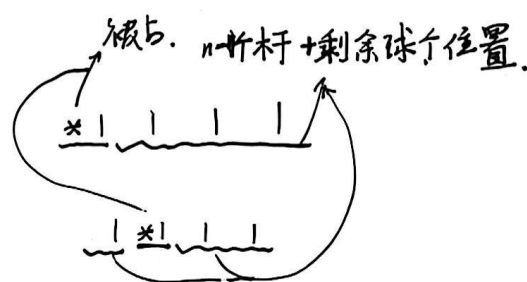
$$p = P(A_1 \cup A_2 \cup \dots \cup A_n) \leq \sum_{i=1}^n \sum_{k=\frac{n}{2}}^n \frac{C_{2n-k-2}^{n-k}}{C_{2n-1}^n} \times$$

(d).

$$p = P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2).$$

$$= \frac{C_{2n-k-2}^{n-k}}{C_{2n-1}^n} \cdot 2 - \frac{1}{C_{2n-1}^n} \times$$

$$= \frac{2C_{2n-k-2}^{n-k} - 1}{C_{2n-1}^n}$$



B_i : i balls are still thrown into the first bin.

(e).

$$p = \sum_{i=0}^{n-1} (P(B_i | \text{first ball's bin}) : \frac{1}{i+1}) \rightarrow \text{独立}.$$

$$= \sum_{i=0}^{n-1} \left(\frac{P(B_i \cap \text{first ball's bin})}{P(\text{first ball's bin})} \cdot \frac{1}{i+1} \right) \times$$

$$= \sum_{i=0}^{n-1} \left(\frac{C_{2n-i-2}^{n-i-1}}{C_{2n-1}^n} \cdot \frac{1}{i+1} \right),$$





7.

(a). $\Omega = \left\{ (v_1, v_2), (v_1, v_3), \dots, (v_1, v_n), (v_2, v_3), \dots \right\}$ $\xrightarrow{C_n^2}$

$$|\Omega| = C_n^2 = 2^{\frac{n(n-1)}{2}}$$

(b)

$$P(E_S) = \frac{1}{C_n^2} = \frac{1}{2^{\frac{n(n-1)}{2}}}$$

(c). ~~??~~ $P(E_{v_1} \cap E_{v_2}) =$

(d).

$$C_n^k = \frac{n!}{(n-k)!k!} = \frac{n \cdot (n-1) \cdots (n-k+1)}{k!} \approx k^k$$

$$= \frac{n \cdot (n-1) \cdots (n-k+1)}{k \cdot (k-1) \cdots 1} \approx k^k$$

(e).

$$= \frac{n}{k} \cdot \frac{n-1}{k-1} \cdots \frac{n-k+1}{1} \leq n^k \text{ 取等条件 } k=1$$

$$p = P(E_{S_1} \cup E_{S_2} \cdots \cup E_{S_{C_n^k}}) \leq C_n^k \cdot P(E_S) = \frac{C_n^k}{2^{\frac{k(k-1)}{2}}} \leq \frac{n^k}{2^{\frac{k(k-1)}{2}}}$$

$$= 2^{k \log n - \frac{k(k-1)}{2}} \leq 2^{-\frac{k(k-1)}{4}} = \frac{1}{2^{\frac{k(k-1)}{4}}} = \frac{1}{2^{\frac{k-1}{4} \cdot k}} \leq \frac{1}{n} \cdot \frac{1}{2^k} \leq \frac{1}{n}$$





2.

① 错误原因: 只做了一半, 问题都没解决.

② 继续改进:

$$\begin{aligned}
 P(\text{target event}) &= P(B | (A \cap B) \cup (\bar{A} \cap \bar{B})) \\
 &= \frac{P(B \cap [(A \cap B) \cup (\bar{A} \cap \bar{B})])}{P[(A \cap B) \cup (\bar{A} \cap \bar{B})]} \\
 &= \frac{P(A \cap B)}{P(A \cap B) + P(\bar{A} \cap \bar{B})} \\
 &= \frac{\frac{8}{1000}}{\frac{8}{1000} + (P(\bar{B}) - P(A \cap \bar{B}))} \\
 &= \frac{\frac{8}{1000}}{\frac{8}{1000} + (\frac{990}{1000} - \frac{891}{1000})} \\
 &= \frac{\frac{8}{1000}}{\frac{8}{1000} + \frac{99}{1000}} = \frac{8}{107} \checkmark
 \end{aligned}$$

③ 答案做法: A 表示测试表明 guilty. B 表示 person 确实是 guilty.

$$\begin{aligned}
 P(B) &= \frac{1}{100} \quad P(\bar{B}) = \frac{99}{100} \quad P(A|B) = \frac{4}{5} \quad P(A|\bar{B}) = \frac{1}{10} \\
 P(B|A) &= \frac{P(B) \cdot P(A|B)}{P(A)} = \frac{P(B) \cdot P(A|B)}{P(B)P(A|B) + P(\bar{B})P(A|\bar{B})} = \frac{8}{107}
 \end{aligned}$$

Bayes's rule.

↓
total.



4. Discussion.

① 错误原因: 有点没理解本题. 按照题解似乎也没有完全模拟好, 因为可能存在二者都不取且的情况, 本题有歧义.

5. ① 错误原因: 事件选取不当, 且没完全弄明白递推. 同时对于事件概率认知不清. → 规模上的.

② 继续改进:

(a) A_i : 第 i 个人拿自己的卡.

$$P(\text{target event}) = P(\bar{A}_1 \cap \bar{A}_2 \cap \dots \cap \bar{A}_n) \\ = P(\overline{A_1 \cup A_2 \cup \dots \cup A_n})$$

sample point 分析

$$= 1 - P(A_1 \cup A_2 \cup \dots \cup A_n)$$

思考: $P(A_i) = \frac{(n-1)!}{n!}$

$$P(A_i \cap A_j) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$$

不要管其他人 就思考第 i 个人拿自己卡的概率

$|S| = n!$ and uniform.

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = \frac{(n-k)!}{n!}$$

$$\therefore \text{上式} = 1 - \left(\sum_{k=1}^n \binom{n}{k} \cdot \frac{(n-k)!}{n!} \right)$$

$$= 1 - \sum_{k=1}^n (-1)^{k-1} \binom{n}{k} \frac{(n-k)!}{n!} = 1 - \sum_{k=1}^n (-1)^{k-1} \frac{1}{k!}$$

$$= - \sum_{k=2}^n (-1)^{k-1} \frac{1}{k!}$$

$$= \sum_{k=2}^n (-1)^k \frac{1}{k!} = \sum_{k=0}^n (-1)^k \frac{1}{k!}$$

$$(-1)^0 \frac{1}{0!} + (-1)^1 \frac{1}{1!} = 0$$

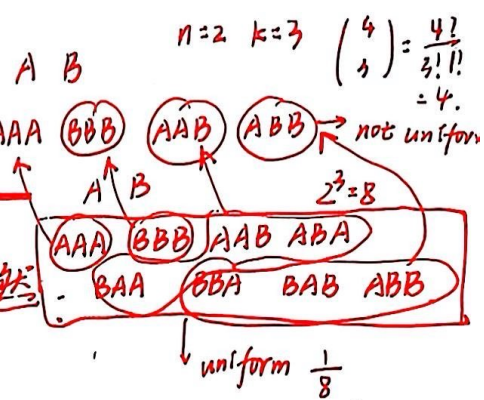
(b).

$$e^{-1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}$$

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③ 答案与图一致.





6. ① 错误原因: 错以为每种无序可重复组合是等概率的, 实则不然.

② 改:

(a).

$|U| = n^n$, uniform.

从 n 个球中选 k 个可重复球为无类. $\binom{n+k-1}{k}$

从 sample points 出.

$$P = \binom{n}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{n-k}$$

(b)

$$P = \sum_{k=\frac{n}{2}}^n \binom{n}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{n-k}$$

A_i : 表示第 i 个箱子有 $\geq \frac{n}{2}$ 个球.

(c).

$P(A_1 \cup A_2 \dots A_n)$

$$P \leq n \cdot \sum_{k=\frac{n}{2}}^n \binom{n}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{n-k}$$

(d).

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$$

$$= 2 \sum_{k=\frac{n}{2}}^n \binom{n}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{n-k} - \frac{\left(\frac{n}{2}\right) \cdot \left(\frac{n}{2}\right) \cdot \left(\frac{n}{2}\right)}{n^n}$$

(e).

B_k : 仍有 k 个球投到第一个依投球箱子里.

$$P(B_k) = \binom{n-1}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{n-k}$$

$$P = \sum_{k=0}^{n-1} \frac{1}{k+1} \binom{n-1}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{n-k-1}$$

③ 答案与②一致.





7. (c).

① 错误原因: 没弄明白.

② 改:

③ 答案:

无约束关系, 完全分开.
 V_1
 边的数量

at most one.

如果 V_1 与 V_2 不共享边, 则 $V_1 \cap V_2 = \emptyset$.

如果 V_1 与 V_2 共享边, 则 $V_1 \cap V_2 = V_3$.

存在约束区域 V_3

$$P(E_{V_1} \cap E_{V_2}) = \sqrt{P(E_{V_1}) P(E_{V_2})}$$

$$P(E_{V_1} \cap E_{V_2}) = \left(\frac{1}{2}\right)^{\binom{|V_1|}{2}} \cdot \left(\frac{1}{2}\right)^{\binom{|V_2|}{2}} - \left(\frac{1}{2}\right)^{\binom{|V_1|}{2} + \binom{|V_2|}{2} - \binom{|V_3|}{2}}$$

$$= \left(\frac{1}{2}\right)^{\binom{|V_1|}{2} + \binom{|V_2|}{2} - \binom{|V_3|}{2}} \neq P(E_{V_1}) \cdot P(E_{V_2})$$
